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Linear differential equations with const co-efficients
$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$

A differential equation of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = 0$$

where $a_0, a_1, a_2, \dots, a_n$ are constant.

If $e^{m_1 x}, e^{m_2 x}, e^{m_3 x}, \dots, e^{m_n x}$ are the solutions of the equation then the general solution is

$$y(x) = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

where $c_1, c_2, c_3, \dots, c_n$ are constant.

Solution procedure:

1 $\rightarrow$ we have to take $y = e^{mx}$ as a solution of the equation.

2. Substituting $y = e^{mx}$ in the given equation then we will get auxiliary equation.

Such as: if $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ (1)

Let $y = e^{mx}$ is solution of (1), then

$$\frac{dy}{dx} = m e^{mx}$$

$$\frac{d^2y}{dx^2} = m^2 e^{mx}$$

$$\therefore (1) \Rightarrow m^2 e^{mx} + 2m e^{mx} + e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 2m + 1) = 0$$

$$\neq e^{mx} \neq 0, m^2 + 2m + 1 = 0$$

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$y \neq 0$ for $e^{mx} = 0$ for a differential equation implies
 or $e^{mx} \neq 0$ for $x \neq 0$

$m^2 + 2m + 1 = 0$, is called an auxiliary equation.

The value of m indicates solution pattern

If $m = \alpha, \beta$, then solution $y(x) = e^{\alpha x} + e^{\beta x}$

$m = \alpha, \alpha$ " " $y(x) = (C_1 + x C_2) e^{\alpha x}$

$m = \pm i\beta$ " " $y(x) = A \cos \beta x + B \sin \beta x$

$m = \alpha \pm i\beta$ " " $y(x) = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$

Problem

$$2 \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + y = 0$$

Solⁿ:

Given that $2 \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + y = 0$ — (1)

Let $y = e^{mx}$ be a solution of (1).

then $\frac{dy}{dx} = m e^{mx}$, $\frac{d^2 y}{dx^2} = m^2 e^{mx}$

$\therefore$ eqn (1) becomes as

$$2m^2 e^{mx} - 3m e^{mx} + e^{mx} = 0$$

$$\Rightarrow (2m^2 - 3m + 1) e^{mx} = 0$$

Since $e^{mx} \neq 0$, $2m^2 - 3m + 1 = 0$

$$\therefore \text{A.E.: } 2m^2 - 3m + 1 = 0 \text{ — (2)}$$

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From eqn (2), $2m^2 - 2m - m + 1 = 0$

$$\Rightarrow 2m(m-1) - 1(m-1) = 0$$

$$\Rightarrow (m-1)(2m-1) = 0$$

$$\Rightarrow m = 1, \frac{1}{2}$$

$\therefore$ The general solution of the given equation is, $y(x) = c_1 e^x + c_2 e^{\frac{1}{2}x}$

$5 \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 3y = 0$ — (1)

Solⁿ: Let $y = e^{mx}$ be a solution of the given equation (1), then $\frac{dy}{dx} = m e^{mx}$, $\frac{d^2 y}{dx^2} = m^2 e^{mx}$
 $\therefore$ Equation (1) can be written as,

$$5m^2 e^{mx} - 2m e^{mx} + 3 e^{mx} = 0$$

$$\Rightarrow (5m^2 - 2m + 3) e^{mx} = 0$$

$$\Rightarrow 5m^2 - 2m + 3 = 0 \quad [\because e^{mx} \neq 0]$$

$$\Rightarrow m = \frac{2 \pm \sqrt{4 - 60}}{2 \cdot 5} = \frac{2 \pm \sqrt{-56}}{10}$$

$$= \frac{2 \pm \sqrt{-56}}{10}$$

$$= \frac{2 \pm 2i\sqrt{14}}{10}$$

$$= \frac{1}{5} \pm \frac{1}{5} i\sqrt{14}$$

$\therefore$ solution of the given equation is

$$y(x) = e^{\frac{1}{5}x} \left(A \cos \frac{\sqrt{14}}{5} x + B \sin \frac{\sqrt{14}}{5} x \right)$$

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Find the particular solutions of

$$\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0 \quad \text{when } y(0) = 0 \text{ and } y'(0) = 1.$$

Soln: ~~Let~~ $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$ — (1)

Let $y = e^{mx}$ be a solution of (1),

$$\text{then } \frac{dy}{dx} = me^{mx} \text{ and } \frac{d^2 y}{dx^2} = m^2 e^{mx}$$

$\therefore$ Equation (1) becomes, $m^2 e^{mx} + 3m e^{mx} + 2e^{mx} = 0$

$\therefore$ Particular solution of the given equation is

$$y(x) = e^{-x} - e^{-2x}$$

$$\Rightarrow (m^2 + 3m + 2)e^{mx} = 0$$

$$\Rightarrow m^2 + 3m + 2 = 0 \quad [\because e^{mx} \neq 0]$$

$$\Rightarrow m^2 + 2m + m + 2 = 0$$

$$\Rightarrow m(m+2) + 1(m+2) = 0$$

$$\Rightarrow (m+2)(m+1) = 0$$

$$\Rightarrow m = -1, -2$$

$\therefore$ The general solution is $y(x) = c_1 e^{-x} + c_2 e^{-2x}$

Since, $y(0) = 0$,

and $y'(0) = 1$,

$$\text{Now } y'(x) = -c_1 e^{-x} - 2c_2 e^{-2x} \quad \text{--- (2)}$$

$$y(0) = c_1 + c_2 = 0$$

$$\Rightarrow c_1 = -c_2$$

$$y'(0) = -c_1 - 2c_2 = 1$$

$$\Rightarrow c_1 + 2c_2 = -1$$

$$\Rightarrow -c_2 + 2c_2 = -1 \Rightarrow c_2 = -1$$

$$\therefore c_1 = 1$$

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Solve: (i) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 0$

(ii) $5\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 3y = 0$

(iii) Find the particular solutions of

$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 8y = 0$ when $y(0) = 0$ & $y'(0) = 8$

(iv) $\frac{d^2y}{dx^2} - 4y = 0$, when $y(0) = 0$ and $y'(0) = 3$.